INCH POUND

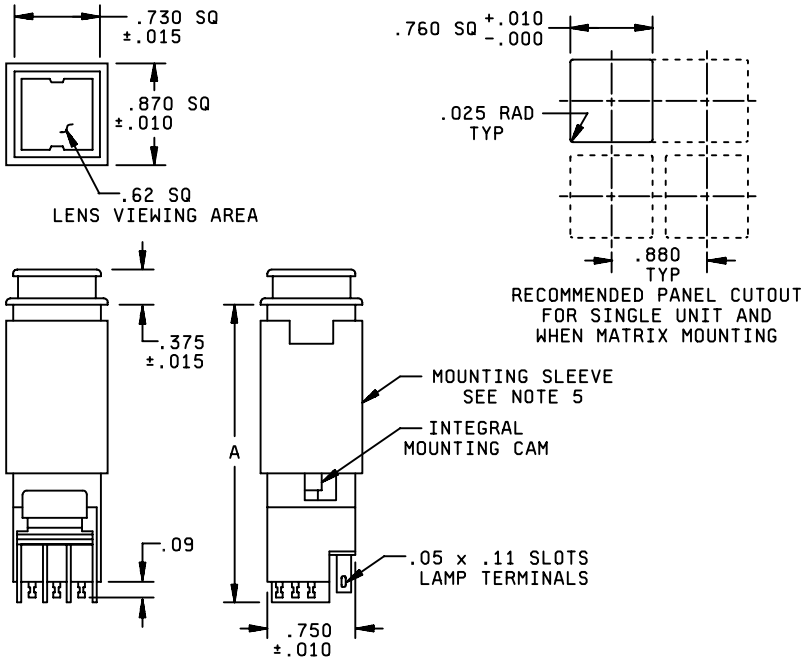
MIL-PRF-22885/75E
1 March 2001
SUPERSEDING
MIL-PRF-22885/75D
25 July 1990

PERFORMANCE SPECIFICATION SHEET

SWITCHES, PUSH BUTTON, ILLUMINATED, 4-LAMP, 0.87 SQUARE, 7 AMPERES, ELECTROMAGNETIC INTERFERENCE SHIELDED

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the indicator lights described herein shall consist of this specification sheet and MIL-PRF-22885.

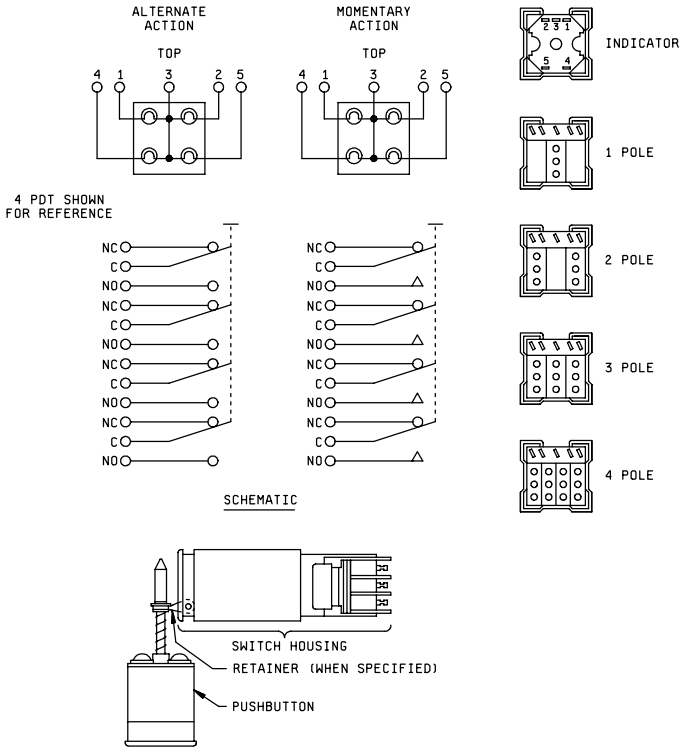


Dimension A	
Momentary	2.750 max
Alternate	2.750 max
Indicator light	1.906 max

FIGURE 1. Design and construction.

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NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals.
3. Metric equivalents are given for general information only.
4. Unit mounts in panels from .125 (3.18 mm) to .250 (6.35 mm) thick.
5. Mounting sleeve shall not exceed .87 square and shall be reversible to accommodate increased panel thickness.
6. Pushbutton designed to prevent incorrect insertion into switch housing.
7. Exact shape of switch optional provided dimensions specified are not exceeded.

FIGURE 1. Design and construction - Continued.

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REQUIREMENTS:

Design and construction: See figure 1. Basic switches shall be MIL-PRF-8805/4 (MS24547-1) or MIL-PRF-8805/109 (M8805/109-01) listed in QPL-8805.

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Vibration grade: 2 (10-500 Hz).

Operation: A (momentary action), B (alternate action), H (indicator light).

Illuminated color: See table I.

Display type:

W (white until illuminated, then background appears in color).

Material:

Housing: Corrosion resistant steel or aluminum alloy.

Weight: 2.2 ounces maximum.

Military part number: See table I.

Lamps: Pushbutton shall accommodate four lamps, T-1 ¾ midget flange base.
Lamps shall be acquired separately.

Operating characteristics:

Actuation force: 2 to 5 pounds.

Pushbutton travel: .218 ±.031.

Lens extraction force: 2 to 8 pounds.

Shock: MIL-STD-202, method 214, test condition H, (75 g's for 6 milliseconds.

Moisture resistance: Following step 6 of the final cycle of the moisture resistance test, while switches are still wet, the insulation resistance measurement shall be not less than 5 megohms.

Electrical endurance: See table II for electrical load conditions.

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Chromaticity coordinates 1/ 2200° kelvin		
Colors	X	Y
Green	.350	SL 2/
	.350	.610
	.185	.610
	.270	SL 2/
Amber	.600	SL 2/
	.588	.385
	.627	.346
	.640	SL 2/
Yellow	.530	.442
	.544	SL 2/
	.571	.402
	.584	SL 2/
Red	.680	.300
	.690	.310
	.723	.257
	.736	.264
Blue	.180	.300
	.180	.370
	.120	.370
	.120	.300
White	.484 ref	.415 ref
X is not less than .350. Y is not greater than .520. $Y - Y_o$ or $Y_o - Y$ is not numerically greater than 0.05X. (Y_o is the Y coordinate of the Planckian radiator for a given X).		

1/ The chromaticities of the diffusing lenses or colored filters expressed as X and Y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using MIL-L-6363/8-7 lamps .30 ±.02 V dc reference 2075°/2100° kelvin.

2/ SL = spectrum locus (where intersected by other of coordinate pair).

TABLE II. Electrical ratings.

Load	Sea level		50,000 feet 28 V dc
	28 V dc	115 V ac, 60 Hz	
	amperes	amperes	amperes
Resistive	7	7	4
Inductive	4	7	2.5
Lamp	2.5	2	2.5

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EMI/RFI shielding: Shielding efficiency test applicable, except the shielding attenuation shall be not less than 47 dB at 100 MHz, 39 dB at 400 MHz and 33 dB at 1,000 MHz.

Luminance: See table III.

TABLE III. Luminance.

Color	Color code	Brightness 1/ 2/ (foot-lamberts, minimum average)
Blue	B	12.0
Green	G	80.0
Red	R	56.0
White	W	146.0
Amber	A	406.0
Yellow	Y	470.0

1/ When illuminated by two M6363/8-7 lamps of 0.34 ±0.02 mean spherical candlepower at 28 volts dc, readings are for lens configuration 0.
2/ Brightness is reduced by approximately 50 percent when EMI screen is specified.

Marking: Lamp circuit schematic shall be marked on the switch housing.

Qualification inspection:

Group submission: All part numbers will be approved for inclusion in the qualified products list upon successful completion of testing specified in table IV. Table IV is based on use of qualified MIL-PRF-8805/4 or MIL-PRF-8805/109 basic switches.

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TABLE IV. Qualification inspection - group submission.

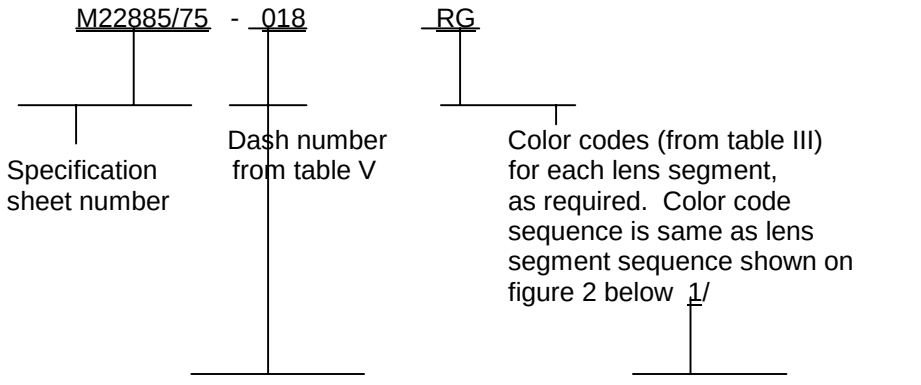
1/ Maintained depressed action.

TABLE II. Illuminated chromaticity limits. 1/

Inspection 1/	Sample
Group I	All samples
Group II	M22885/75-201R (4 samples)
Group III	M22885/75-025W (2 samples)
Group VI 2/	M22885/75-025W (2 samples)
Group VII	M22885/75-201R (2 samples)
Group VIII	2 pushbuttons of each color (10 samples), of lens configuration 0, and without EMI/RFI shielding. 3/

1/ Inspection groups in accordance with qualification inspection table of MIL-PRF-22885.
2/ Electrical endurance: Test resistance load, ac, only. Lamps used during electrical endurance test shall be M6363/8-7.
3/ Pushbuttons shall be assembled to any representative switch body.

Part numbers: Part numbers are assigned as illustrated below. Part numbers do not include legend description. Acquire switches with legends as category II items in accordance with MIL-PRF-22885.



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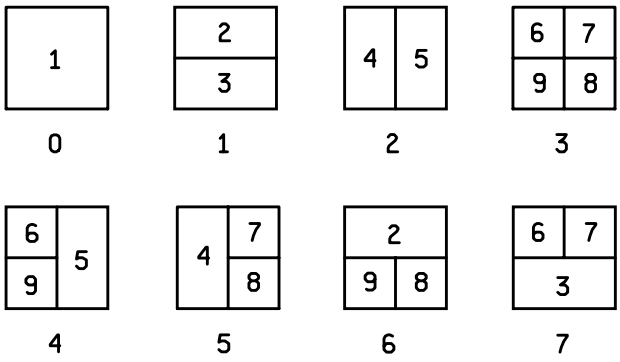


FIGURE 2. Lens configuration and segment numbers.

1/ Part number constructed as shown replace and are interchangeable with previous part numbers which included segment numbers in addition to color codes (example: M22885/75-018-2R3G).

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TABLE V. Dash numbers.

Dash numbers		Operation	Circuit	Lens configuration (see figure 2)	Retained pushbutton 1/
Non-EMI shielded	EMI shielded				
001	---	Momentary	1PDT	0	Yes
002	---	Momentary	1PDT	1	Yes
003	---	Momentary	1PDT	2	Yes
2/ 004	---	Momentary	1PDT	3	Yes
005	---	Momentary	1PDT	4	Yes
006	---	Momentary	1PDT	5	Yes
007	---	Momentary	1PDT	6	Yes
008	---	Momentary	1PDT	7	Yes
009	161	Momentary	2PDT	0	Yes
010	162	Momentary	2PDT	1	Yes
011	163	Momentary	2PDT	2	Yes
012	164	Momentary	2PDT	3	Yes
013	165	Momentary	2PDT	4	Yes
014	166	Momentary	2PDT	5	Yes
015	167	Momentary	2PDT	6	Yes
016	168	Momentary	2PDT	7	Yes
017	169	Momentary	3PDT	0	Yes
018	170	Momentary	3PDT	1	Yes
019	171	Momentary	3PDT	2	Yes
020	172	Momentary	3PDT	3	Yes
021	173	Momentary	3PDT	4	Yes
022	174	Momentary	3PDT	5	Yes
023	175	Momentary	3PDT	6	Yes
024	176	Momentary	3PDT	7	Yes
025	177	Momentary	4PDT	0	Yes
026	178	Momentary	4PDT	1	Yes
027	179	Momentary	4PDT	2	Yes
028	180	Momentary	4PDT	3	Yes
029	181	Momentary	4PDT	4	Yes
030	182	Momentary	4PDT	5	Yes
031	183	Momentary	4PDT	6	Yes
032	184	Momentary	4PDT	7	Yes
033	---	Alternate	1PDT	0	Yes
034	---	Alternate	1PDT	1	Yes
035	---	Alternate	1PDT	2	Yes
2/ 036	---	Alternate	1PDT	3	Yes
037	---	Alternate	1PDT	4	Yes
038	---	Alternate	1PDT	5	Yes
039	---	Alternate	1PDT	6	Yes
040	---	Alternate	1PDT	7	Yes
041	185	Alternate	2PDT	0	Yes
042	186	Alternate	2PDT	1	Yes
043	187	Alternate	2PDT	2	Yes
044	188	Alternate	2PDT	3	Yes
045	189	Alternate	2PDT	4	Yes
046	190	Alternate	2PDT	5	Yes
047	191	Alternate	2PDT	6	Yes
048	192	Alternate	2PDT	7	Yes

See footnotes at end of table.

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TABLE V. Dash numbers - Continued.

Dash numbers		Operation	Circuit	Lens configuration (see figure 2)	Retained pushbutton 1/
Non-EMI shielded	EMI shielded				
049	193	Alternate	3PDT	0	Yes
050	194	Alternate	3PDT	1	Yes
051	195	Alternate	3PDT	2	Yes
052	196	Alternate	3PDT	3	Yes
053	197	Alternate	3PDT	4	Yes
054	198	Alternate	3PDT	5	Yes
055	199	Alternate	3PDT	6	Yes
056	200	Alternate	3PDT	7	Yes
057	201	Alternate	4PDT	0	Yes
058	202	Alternate	4PDT	1	Yes
059	203	Alternate	4PDT	2	Yes
060	204	Alternate	4PDT	3	Yes
061	205	Alternate	4PDT	4	Yes
062	206	Alternate	4PDT	5	Yes
063	207	Alternate	4PDT	6	Yes
064	208	Alternate	4PDT	7	Yes
209	217	Indicator	Indicator	0	Yes
210	218	Indicator	Indicator	1	Yes
211	219	Indicator	Indicator	2	Yes
212	220	Indicator	Indicator	3	Yes
213	221	Indicator	Indicator	4	Yes
214	222	Indicator	Indicator	5	Yes
215	223	Indicator	Indicator	6	Yes
216	224	Indicator	Indicator	7	Yes
2/ 065	113	Momentary	2PDT	0	No
066	114	Momentary	2PDT	1	No
067	115	Momentary	2PDT	2	No
068	116	Momentary	2PDT	3	No
069	117	Momentary	2PDT	4	No
070	118	Momentary	2PDT	5	No
071	119	Momentary	2PDT	6	No
072	120	Momentary	2PDT	7	No
2/ 073	121	Momentary	3PDT	0	No
074	122	Momentary	3PDT	1	No
075	123	Momentary	3PDT	2	No
076	124	Momentary	3PDT	3	No
077	125	Momentary	3PDT	4	No
078	126	Momentary	3PDT	5	No
079	127	Momentary	3PDT	6	No
080	128	Momentary	3PDT	7	No
2/ 081	129	Momentary	4PDT	0	No
082	130	Momentary	4PDT	1	No
083	131	Momentary	4PDT	2	No
084	132	Momentary	4PDT	3	No
085	133	Momentary	4PDT	4	No
086	134	Momentary	4PDT	5	No
087	135	Momentary	4PDT	6	No
088	136	Momentary	4PDT	7	No

See footnotes at end of table.

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TABLE V. Dash numbers - Continued.

Dash numbers		Operation	Circuit	Lens configuration (see figure 2)	Retained pushbutton 1/
Non-EMI shielded	EMI shielded				
2/ 089	2/ 137	Alternate	2PDT	0	No
090	138	Alternate	2PDT	1	No
091	139	Alternate	2PDT	2	No
092	140	Alternate	2PDT	3	No
093	141	Alternate	2PDT	4	No
094	142	Alternate	2PDT	5	No
095	143	Alternate	2PDT	6	No
096	144	Alternate	2PDT	7	No
2/ 097	2/ 145	Alternate	3PDT	0	No
098	146	Alternate	3PDT	1	No
099	147	Alternate	3PDT	2	No
100	148	Alternate	3PDT	3	No
101	149	Alternate	3PDT	4	No
102	150	Alternate	3PDT	5	No
103	151	Alternate	3PDT	6	No
104	152	Alternate	3PDT	7	No
2/ 097	2/ 153	Alternate	4PDT	0	No
098	098	Alternate	4PDT	1	No
099	099	Alternate	4PDT	2	No
100	100	Alternate	4PDT	3	No
101	101	Alternate	4PDT	4	No
102	102	Alternate	4PDT	5	No
103	103	Alternate	4PDT	6	No
104	104	Alternate	4PDT	7	No

1/ Retained pushbutton shall be capable of having lamps replaced but is restricted in complete removal from switch housing.
2/ Not for new design.

NOTE: 800 and 900 series dash numbers, for switch housing only and pushbutton only, have been canceled without replacement.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, AV, CR4
Navy - AS, MC, OS, SH
Air Force - 99

(Project 5930-1716-10)

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